Combined Conditional Probabilities

Exam code: 0607

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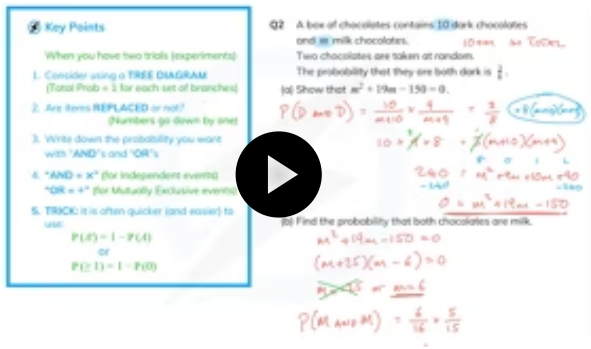

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Combined conditional probabilities

What is a combined conditional probability?

- This is when you have two (or more) **successive events**, one after the other, and the **second** event **depends on** (is conditional on) the **first**

How do I calculate combined conditional probabilities?

- You need to **adjust** the number of outcomes as you go along
  - For example, selecting two cards from a pack of 52 playing cards **without replacing** the first card:
    - P(red 1st card) is 26 reds out of 52 cards
    - If the 1st card is not replaced, there are only 25 reds left out the remaining 51 cards
    - P(red 2nd card) is 25 reds out of 51 cards
    - P(red then red) = $\frac{26}{52} \times \frac{25}{51}$

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

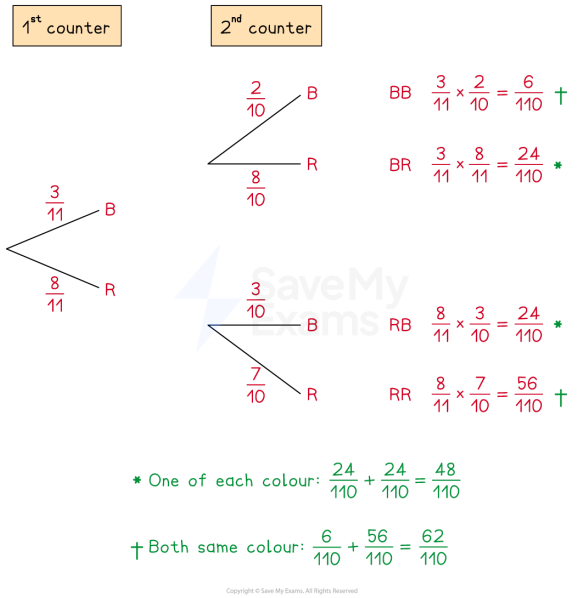
Got it

Examiner Tips and Tricks

If a question says "two cards are drawn" then you may **assume** that they draw 1 card followed by another card **without replacement** (the maths is the same).

Can I draw a tree diagram for combined conditional probabilities?

- Yes, a **tree diagram** is a useful way to show combined conditional probabilities
  - For example, two counters are drawn at random from a bag of 3 blue and 8 red counters without replacement
    - The probabilities are shown below



What if there are multiple possibilities within one question?

- You may need a **listing strategy** (e.g. *AAB*, *ABA*, *BAA*)
- You will need the **or** rule for multiple possibilities
  - P(*AB* **or** *BA* **or** *AA* **or**...) = P(*AB*) + P(*BA*) + P(*AA*) +...
    - **Add** the cases together

- Remember that *AB* and *BA* are **not the same**
  - *AB* means *A* happened first, then *B*
  - *BA* means *B* happened first, then *A*

Examiner Tips and Tricks

Worked Example

A bag contains 10 yellow beads, 6 blue beads and 4 green beads.

A bead is taken at random from the bag and not replaced.

A second bead is then taken at random from the bag.

(a) Find the probability that both beads are different colours.

Answer:

Let Y, B and G represent choosing a yellow, blue and green bead

Method 1

The probability of the beads being different colours is equal to 1 minus the probability that the beads are the same colour

Find the probability of both beads being the same colour

P(same colours) = P(YY) + P(BB) + P(GG)

Calculate each conditional probability separately, remembering the number of beads changes after one is drawn and not replaced

For example, P(YY) = $\frac{10}{20} \times \frac{9}{19}$

$$\frac{10}{20} \times \frac{9}{19} + \frac{6}{20} \times \frac{5}{19} + \frac{4}{20} \times \frac{3}{19}$$

Multiply the pairs of fractions together and add their results

$$\frac{232}{380}$$

Subtract this from 1

$$1 - \frac{132}{380} = \frac{248}{380}$$

Simplify the answer

$$\frac{62}{95}$$

Method 2

List all the possibilities of different colours

Remember that YB (yellow first, then blue) is different to BY (blue first, then yellow)

YB, BY, YG, GY, BG, GB

Use the "or" rule to add the cases together

P(different colours) = P(YB) + P(BY) + P(YG) + P(GY) + P(BG) + P(GB)

For example, P(YB) = $\frac{10}{20} \times \frac{6}{19}$

$$\frac{10}{20} \times \frac{6}{19} + \frac{6}{20} \times \frac{10}{19} + \frac{10}{20} \times \frac{4}{19} + \frac{4}{20} \times \frac{10}{19} + \frac{6}{20} \times \frac{4}{19} + \frac{4}{20} \times \frac{6}{1}$$

Multiply the pairs of fractions together and add their results

$$\frac{248}{380}$$

Simplify the answer

$$\frac{62}{95}$$

(b) The second bead is not replaced and a third bead is taken at random from the bag.

Find the probability that all three beads are the same colour.

Answer:

List the possibilities

YYY, BBB, GGG

Use the "or" rule to add between cases

P(all the same colour) = P(YYY) + P(BBB) + P(GGG)

Use conditional probabilities in each separate case, remembering the number of beads changes after each one is drawn and not replaced

$$\frac{10}{20} \times \frac{9}{19} \times \frac{8}{18} + \frac{6}{20} \times \frac{5}{19} \times \frac{4}{18} + \frac{4}{20} \times \frac{3}{19} \times \frac{2}{18}$$

Multiply the triplets of fractions together then add their results

$$\frac{720}{6840} + \frac{120}{6840} + \frac{24}{6840} = \frac{864}{6840}$$

Simplify the answer

$$\frac{12}{95}$$

Worked Example

A bag contains only red and blue marbles. There are 3 more red marbles than blue marbles.

Kai takes 2 marbles from the bag at random.

The probability that Kai takes 2 blue marbles is $\frac{1}{7}$.

Work out the number of blue marbles there were in the bag initially.

Write expressions for the number of each colour

Let x be the number of blue marbles

Then there are $x + 3$ red marbles

And there are $x + x + 3 = 2x + 3$ marbles in total

Find the probability that the first marble is blue

$$\frac{x}{2x+3}$$

Given that the first marble is blue, find the probability that the second marble is blue

- After one blue marble is taken, there are $x - 1$ blue marbles left out of $2x + 2$ marbles in total

$$\frac{x-1}{2x+2}$$

Find the probability that both marbles are blue by multiplying the probabilities together

- Set this equal to $\frac{1}{7}$

$$\frac{x}{2x+3} \times \frac{x-1}{2x+2} = \frac{1}{7}$$

Multiply through by $7(2x+3)(2x+2)$ to get rid of the fractions

$$7x(x-1) = (2x+3)(2x+2)$$

Expand and simplify both sides

$$7x^2 - 7x = 4x^2 + 4x + 6x + 6$$
$$7x^2 - 7x = 4x^2 + 10x + 6$$

Rearrange to make one side equal to zero

$$3x^2 - 17x - 6 = 0$$

Factorise the quadratic

- Find two numbers that multiply to give 3×-6=-18 and add to give -17
- 1 and -18

$$3x^2 + x - 18x - 6$$
$$x(3x+1) - 6(3x+1)$$
$$(3x+1)(x-6)$$

Solve the quadratic equation

$$(3x+1)(x-6) = 0$$
$$x = -\frac{1}{3}, x = 6$$

x needs to be a positive whole number

$x = 6$ **blue marbles initially**

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